Transgenic animals

Transgenic animals are organisms whose genome has been intentionally altered by adding foreign DNA (a transgene) using techniques like microinjection or stem cell transfer to express new, heritable traits.

These animals are used extensively in biomedical research to study human disease, develop drugs, and improve agricultural efficiency, such as disease-resistant livestock.

Key Aspects and Examples:

- **Medical Research (Mice):** Mice are the most used model to study cancer, diabetes, Alzheimer's, and Parkinson’s diseases.

- **Agriculture & Food:** Transgenic cows can produce healthier milk with human proteins (alpha-lactalbumin)

- **Drug Production:** Transgenic sheep and goats have been used to produce human pharmaceuticals, such as proteins, in their milk.

- **Examples:** The first successful transgenic animal was a mouse (1982). Others include pigs (transplants/higher omega-3), fish (GloFish), and sheep.

Transgenic animals are generated by introducing foreign DNA into an animal's genome to modify its traits, largely through:

- **Pronuclear microinjection**
- **ES cell-mediated gene transfer**
- **Retroviral vectors**
- **Somatic cell nuclear transfer (SCNT)**
- **Sperm-Mediated Gene Transfer**

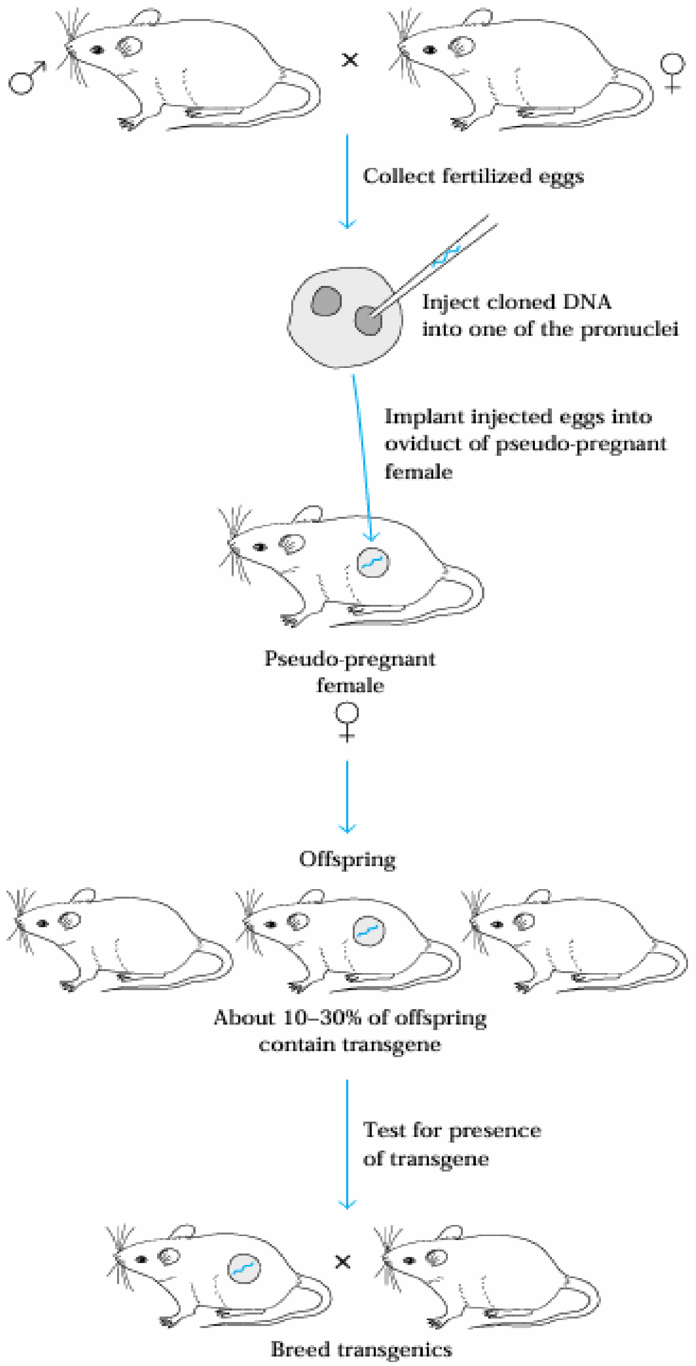
These methods create either gain-of-function (addition) or loss-of-function (knockout) modifications to study diseases or improve agriculture.

Pronuclear Microinjection

•**Technique:** A transgene, typically a linearized DNA fragment (plasmid or BAC-based), is injected into one of the two pronuclei (usually the larger male pronucleus) in a one-cell embryo (zygote).

•**Procedure:**

- **Preparation:** DNA is purified, linearized, and suspended in an injection buffer.
- **Microinjection:** Under a microscope, an embryo is held by a holding pipette while a microinjection needle injects picoliters of DNA solution into the nucleus.
- **Transfer:** Surviving embryos are transferred to the oviduct of a surrogate mother.



General procedure for producing transgenic mice by pronuclear microinjection:

Fertilized eggs are collected from a pregnant female mouse. Cloned DNA (referred to as the transgene) is microinjected into one of the pronuclei of a fertilized egg.

The eggs are then implanted into the oviduct of pseudopregnant foster mothers (obtained by mating normal females with a sterile male).

The transgene will be incorporated into the chromosomal DNA of about 10%–30% of the offspring and will be expressed in all of their somatic cells.

If a tissue-specific promoter is linked to a transgene, then tissue-specific expression of the transgene will result

•**Applications:**

•**Transgenic Models:** Generating transgenic mice to study gene function, disease mechanisms (e.g., Alzheimer's, cancer), and developmental biology.

•**Gene Targeting:** Combining with technologies like CRISPR-Cas9 for precise knockout or knockin modifications.

•**Transgenesis:** Used to introduce foreign genetic information into mammalian genomes.

•**Limitations:**

•**Random Integration:** The transgene typically integrates into the genome randomly.

•**Varying Expression:** The transgene's expression level can vary due to insertion location.

•**Technical Skill:** Requires significant expertise and specialized equipment for success.

ES Cell–Mediated Gene Transfer

Embryonic stem (ES) cell–mediated gene transfer is a highly precise method used to generate **transgenic animals**, particularly for **gene targeting (knockout/knock-in)** studies.

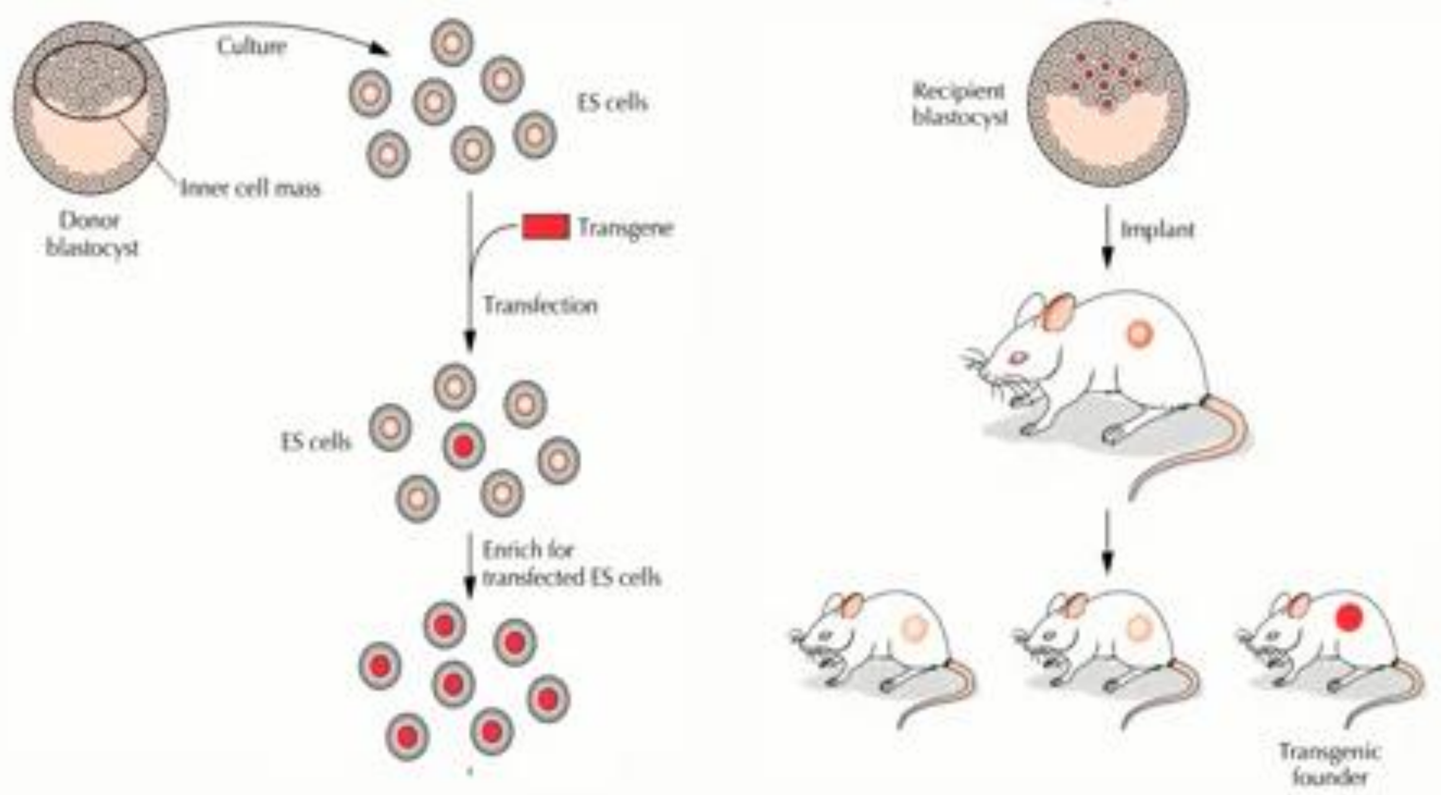
Principle

ES cells are **pluripotent cells** derived from the inner cell mass of a blastocyst. These cells can:

- Self-renew in culture
- Differentiate into all cell types, including germ cells

Genetic modification is introduced into ES cells using Homologous Recombination, allowing **site-specific changes in the genome**.

Embryonic Stem Cell-mediated Gene Transfer



Procedure:

1. Isolation and Culture of ES Cells

- ES cells are isolated from the **blastocyst-stage embryo**
- Maintained in culture under conditions preserving pluripotency

2. Construction of Targeting Vector

- A **DNA construct** is designed containing:
  - Desired gene modification (insertion, deletion, or mutation)
  - Homology arms (for recombination)
  - Selectable marker (e.g., neomycin resistance)

3. Introduction of DNA into ES Cells

- DNA is introduced using techniques like:
  - Electroporation
- The construct integrates into the genome via **Homologous Recombination**.

5. Injection into Blastocyst
Genetically modified ES cells are injected into a recipient blastocyst.

6. Implantation into Surrogate Mother
The blastocyst is implanted into a pseudopregnant female.

7. Formation of Chimeric Animals
Offspring are **chimeras**, containing both host and modified ES cells.

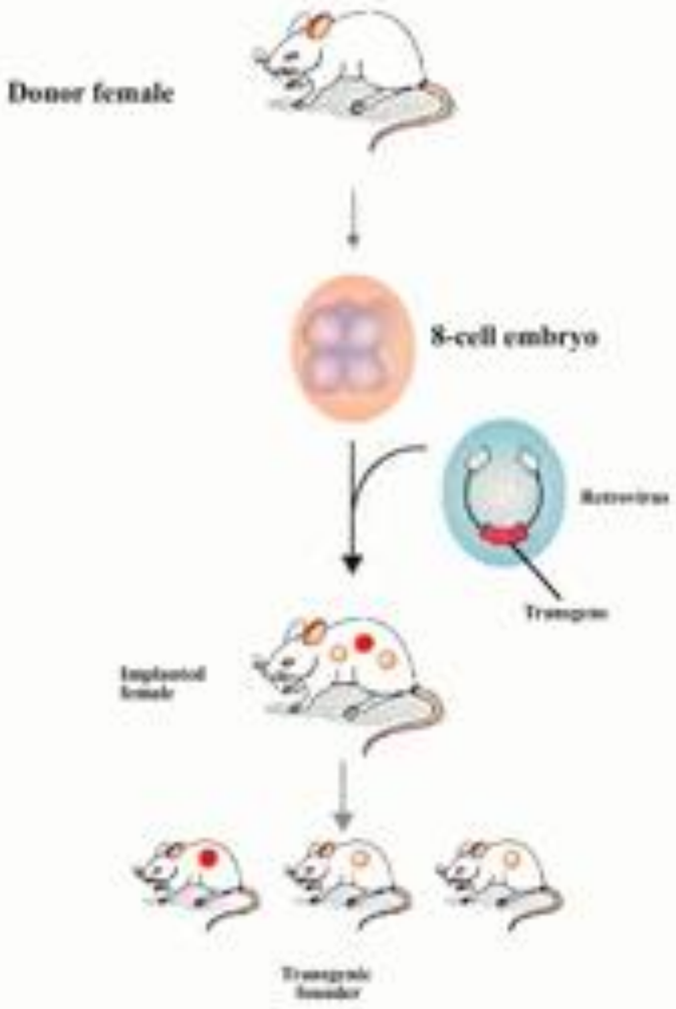
Advantages:

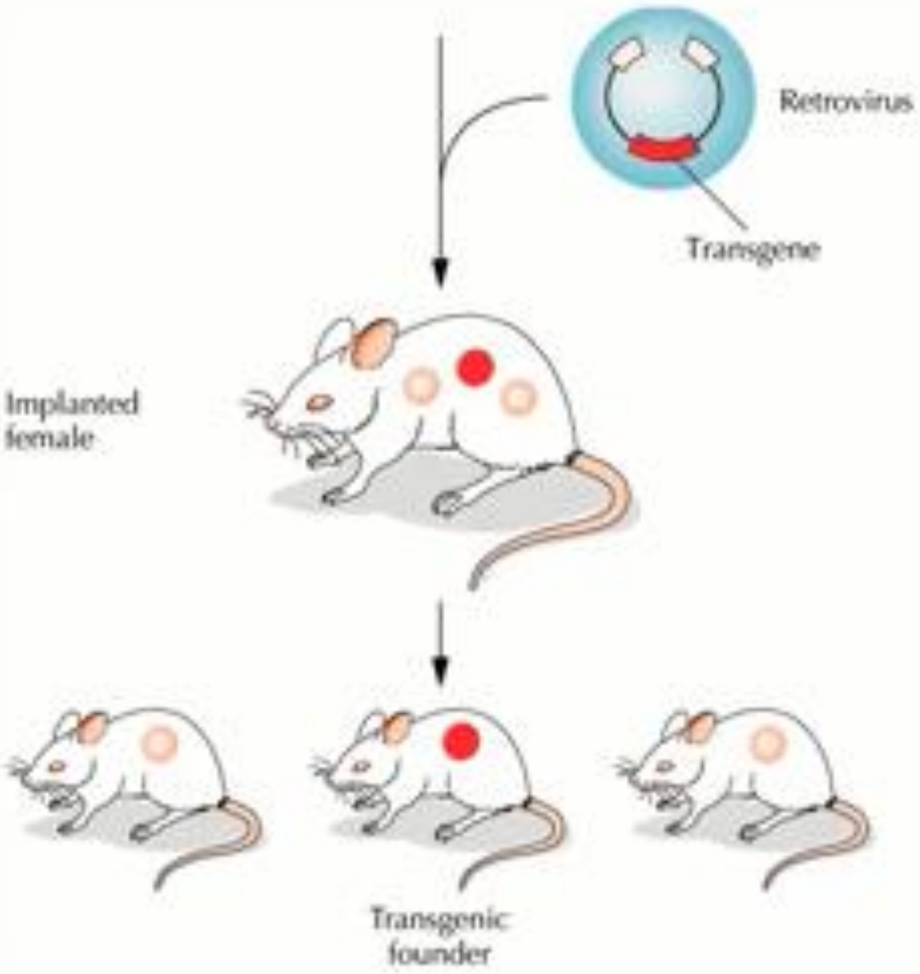
- **High precision:** Enables targeted gene insertion or deletion via homologous recombination.
- **Reliable screening:** Only correctly modified ES cells are selected before embryo introduction.
- **Stable inheritance:** Ensures germline transmission of the modified gene.
- **Versatility:** Allows complex modifications like knockouts and conditional mutations.
- **Reduced randomness:** Minimizes random integration compared to viral methods.

Disadvantages:

- **Time-consuming:** Involves multiple lengthy steps from cell culture to breeding.
- **Technically demanding:** Requires expertise in ES cell handling and microinjection/electroporation.
- **Chimerism:** Produces chimeric animals that need further breeding.
- **Limited species use:** Efficient mainly in mice.

Retroviral Method





Gamma-Retroviral Vectors

Figure 1: Wild-type gamma-retrovirus genome.

In order to produce gamma-retroviral vectors, you need three plasmids (Figure 2):

•**Transfer plasmid** — contains transgene flanked by LTRs; ~8 kb packaging capacity

•**Packaging plasmid** — contains packaging genes *gag* and *pol*

•**Envelope plasmid** — contains packaging gene *env*; usually VSV-G due to wide infectivity

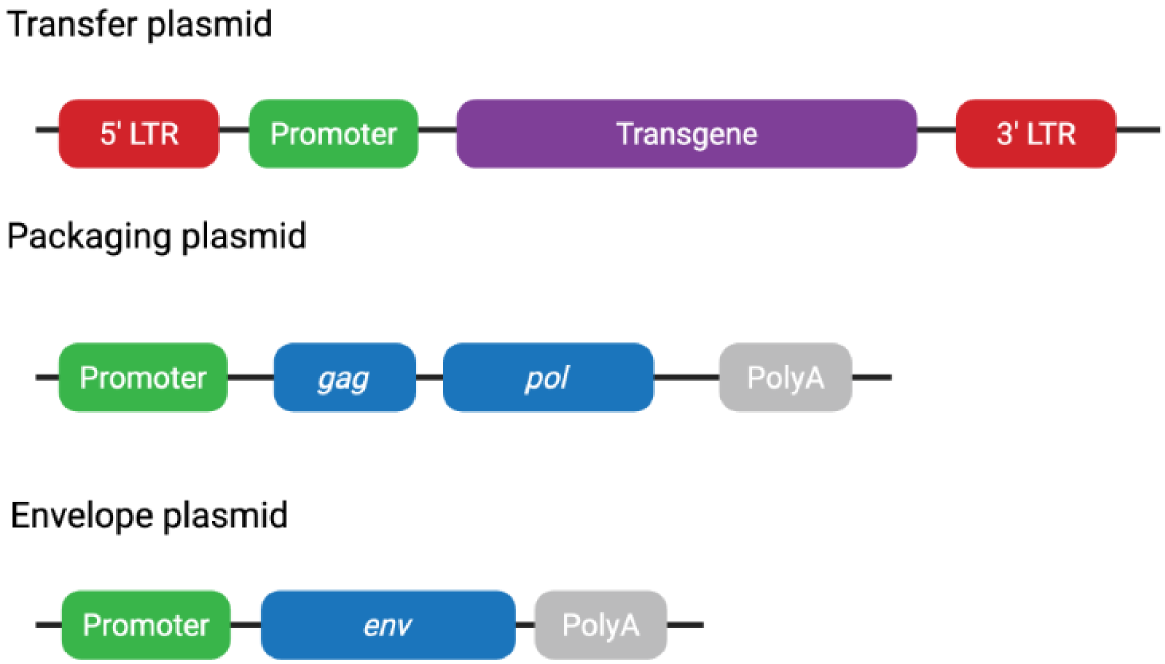


Figure 2: Gamma-retroviral plasmids.

Gamma-Retroviral Vector Production

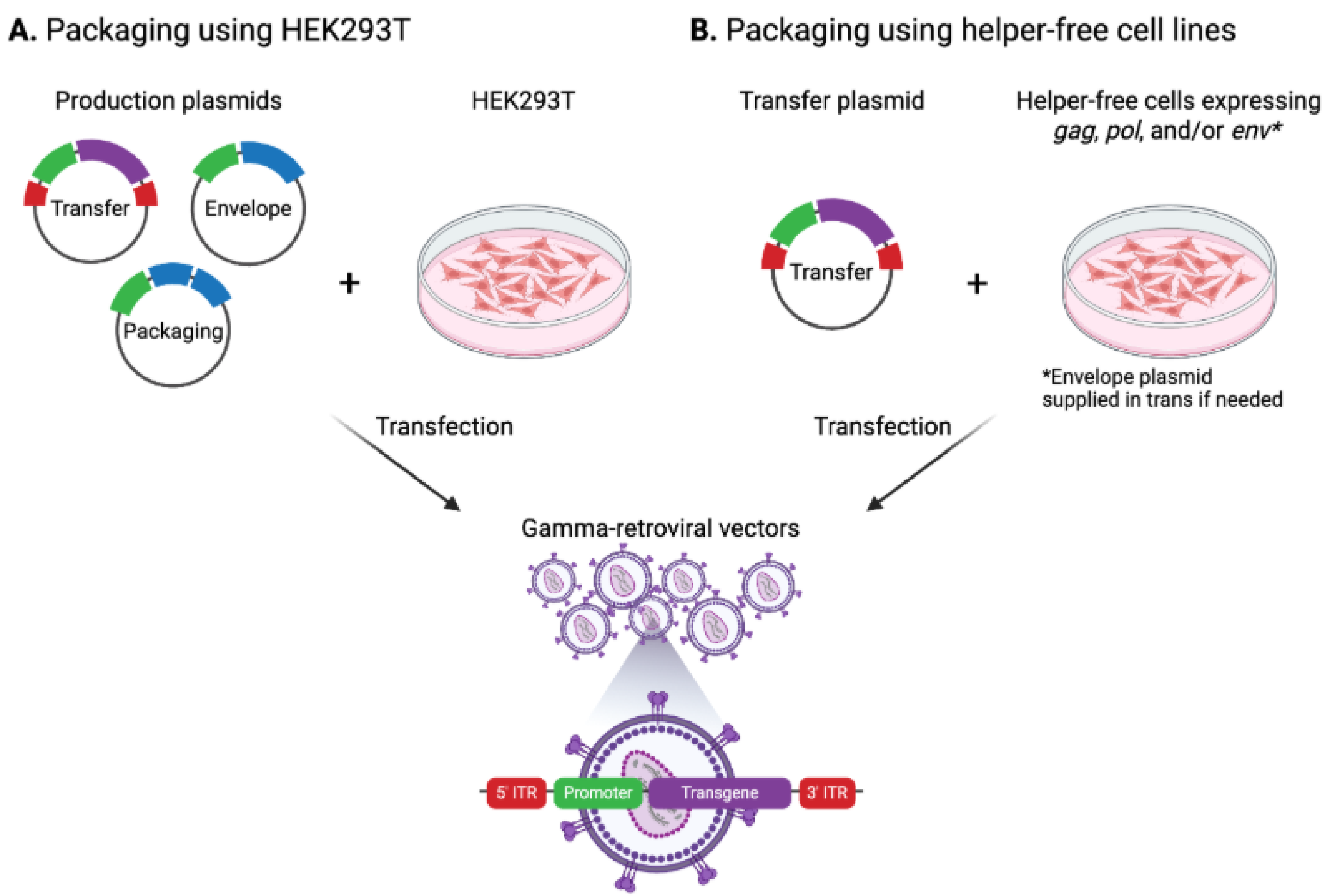


Figure 3: Gamma-retroviral vector packaging methods.

Procedure:

1. Construction of Recombinant Retroviral Vector

•The gene of interest is inserted into a retroviral genome

•Viral genes required for replication are removed (to ensure safety)

2. Packaging of Virus

•Recombinant DNA is introduced into **packaging cell lines**

•These cells produce infectious but replication-deficient viral particles

3. Infection of Early Embryos

•Early-stage embryos (typically 4–8 cell stage) are exposed to the virus

•Virus infects embryonic cells and delivers the transgene

4. Integration into Host Genome

- Viral RNA is converted to DNA and integrates into host genome
- Integration is usually **random**

5. Embryo Transfer

- Infected embryos are implanted into a pseudopregnant female

6. Screening of Offspring

- Offspring are analyzed for presence of the transgene
- Transgenic mice are bred for stable inheritance

Retroviral Mediated Gene Transfer

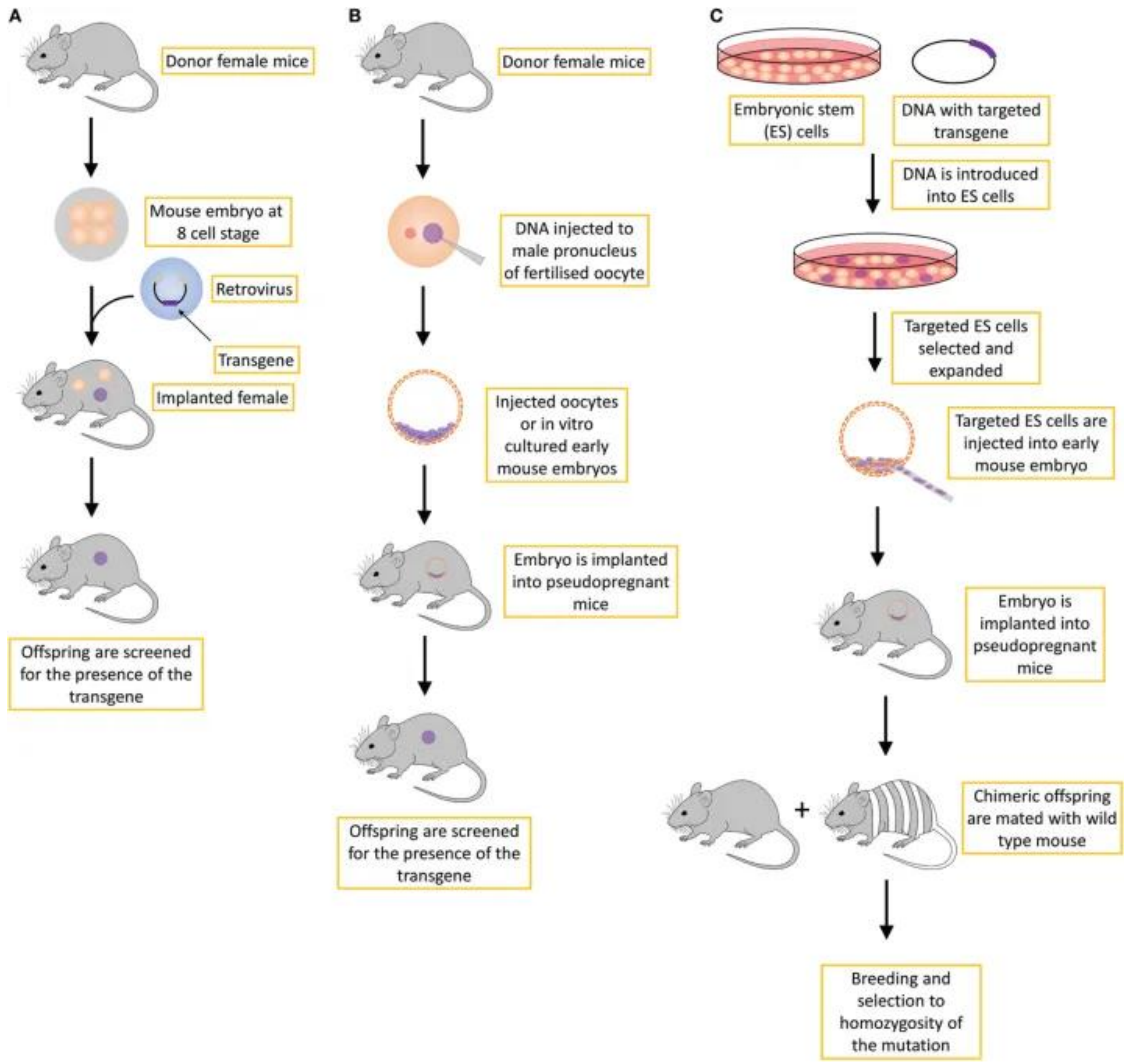
Advantages

- It infects early stage embryo with replication- defective virus.
- The method allows small DNA inserts upto < 8 kb possible.

Disadvantages:

- Random insertion of the transgene occurs.
- Retroviral contamination of the transgenic animal.

Comparison of the three methods



Cre-loxP system

The Cre-loxP system is a powerful site-specific recombination tool derived from P1 bacteriophage, widely used for genome engineering, particularly in generating tissue-specific or inducible conditional knockout mice. It utilizes the <u>Cre recombinase</u> enzyme to recognize and catalyze DNA rearrangement between two 34-base pair *loxP* sites.

Key Components and Mechanisms

•**Cre Recombinase:** A 38-kDa recombinase enzyme.

•**loxP Sites:** Specific 34-bp recognition sequences consisting of two 13-bp palindromic repeats and an 8-bp core spacer.

•**"Floxed" DNA:** A target gene flanked by *loxP* sites (loxP-flanked = floxed).

•**Recombination Types:**

- •**Deletion (Excision):** When *loxP* sites are oriented in the same direction, Cre removes the intervening DNA.
- •**Inversion:** When *loxP* sites are oriented in opposite directions, Cre flips the intervening DNA sequence.
- •**Translocation:** When *loxP* sites are on different chromosomes, Cre can cause translocation.

Cre recombinase, originally named because it "causes recombination" (although later referred to as the "cyclization recombinase"), is a 38 kDa protein responsible for intra- and inter-molecular recombination at the *loxP* recognition sites.

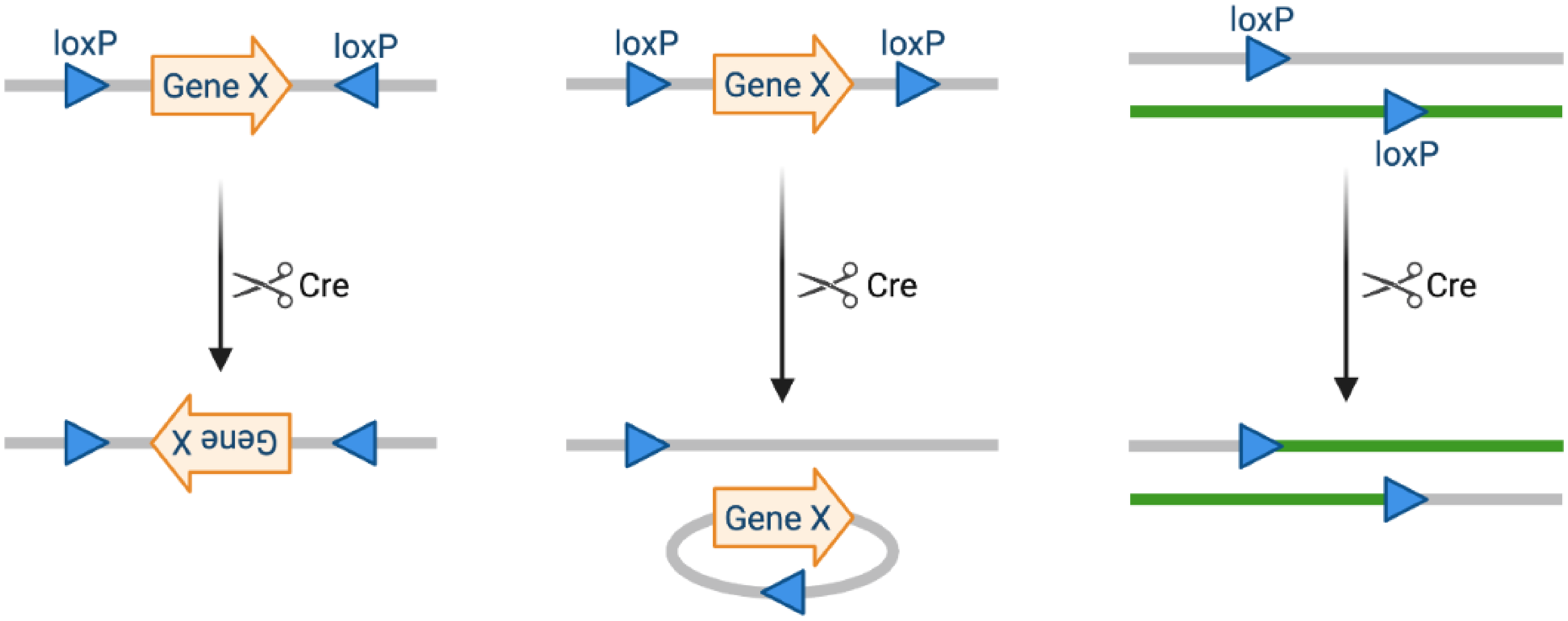
A key advantage of the system is that Cre acts independently of any other accessory proteins or co-factors, thus allowing for broad applications in a variety of experiments.

LoxP (locus of X(cross)-over in P1) sites are 34-base-pair long recognition sequences consisting of two 13-bp long palindromic repeats separated by an 8-bp long asymmetric core spacer sequence.

The asymmetry in the core sequence gives the *loxP* site directionality, and the canonical *loxP* sequence is ATAACTTCGTATA-GCATACAT-TATACGAAGTTAT.

The *loxP* sequence does not occur naturally in any known genome other than P1 phage, and is long enough that there is virtually no chance of it occurring randomly.

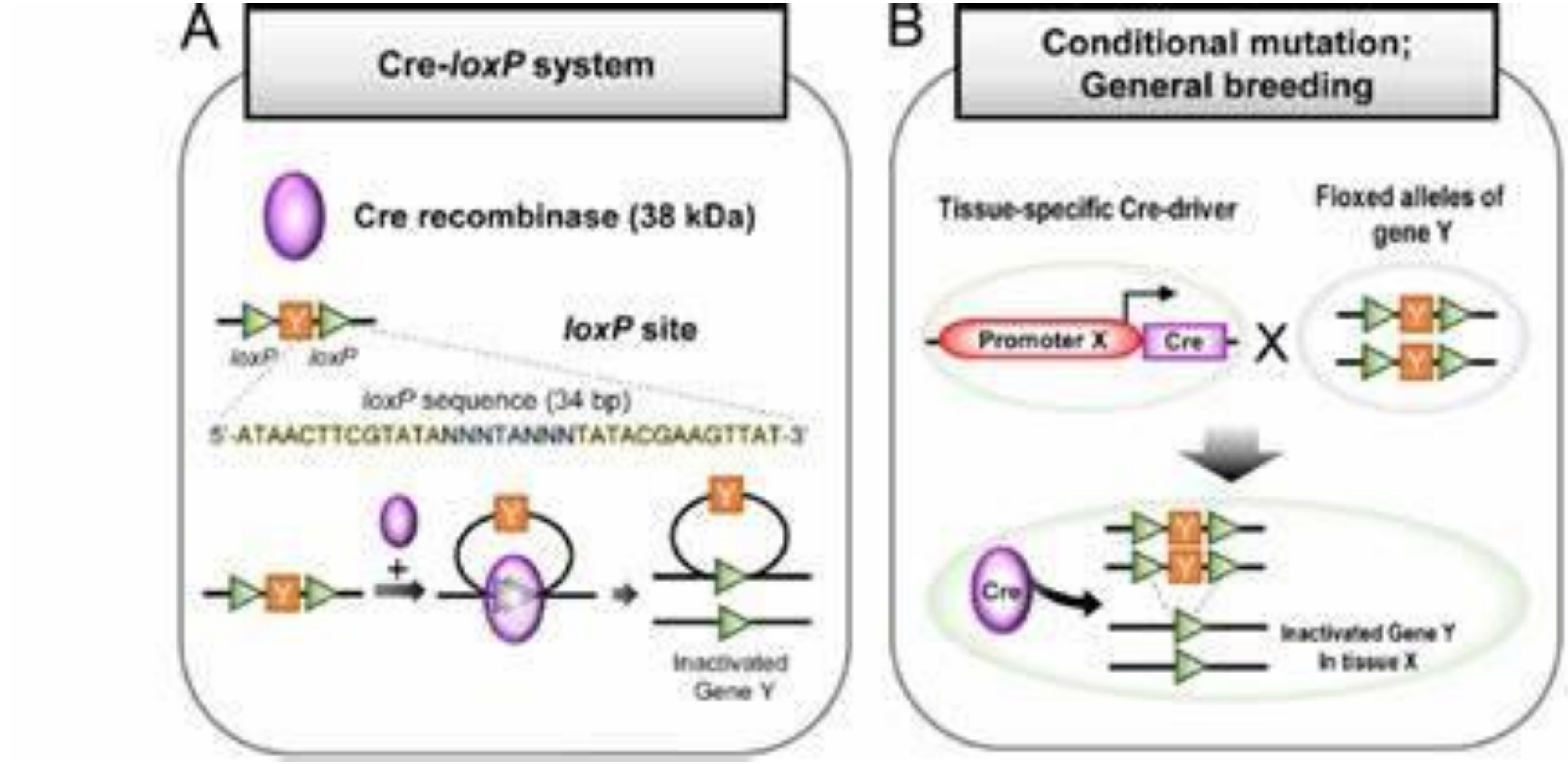
Therefore, inserting *loxP* sites at deliberate locations in a DNA sequence allows for very specific manipulations.



Inversion: If the loxP sites are on the same DNA strand and are in opposite orientations, recombination results in an inversion and the region of DNA between the loxP sites is reversed.

Deletion: If the sites face in the same direction, the sequence between the loxP sites is excised as a circular piece of DNA (and is not maintained).

Translocation: If the sites are on separate DNA molecules, a translocation event is generated at the loxP sites.


A
Cre-loxP system
Cre recombinase (38 kDa)
loxP site
loxP
loxP
loxP sequence (34 bp)
5'-ATAACTTCGTATANNNTANNNTATACGAAGTTAT-3'
Y
+
Inactivated Gene Y
B
Conditional mutation; General breeding
Tissue-specific Cre-driver
Floxed alleles of gene Y
Promoter X
Cre
X
Cre
Inactivated Gene Y In tissue X

Knockout Mice

- Transgenic mice that carry a disrupted gene, called knockout mice, have been extremely helpful to immunologists/cancer biologists trying to understand how the removal of a particular gene product affects the immune system.

- Various knockout mice are being used in immunologic/cancer research.

- Gene-Targeted Knockout Mice Assess the Contribution of a Particular Gene

Gene targeting with Cre/loxP:

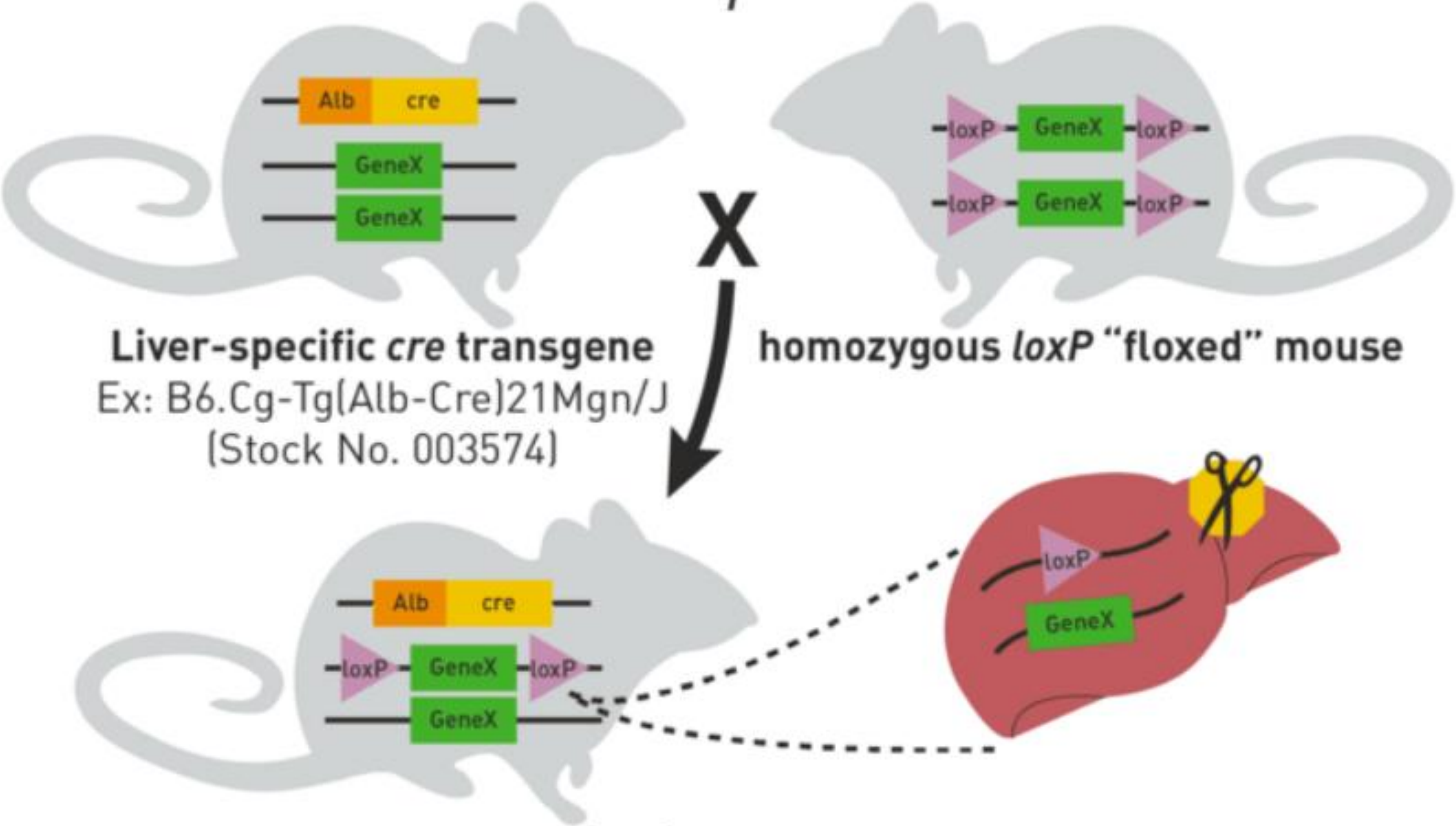
Conditional deletion by Cre recombinase. The targeted gene is modified by flanking the gene with loxP sites.

Mice are generated from ES cells by standard procedures. Mating of the loxP-modified-mice with a Cre transgenic will generate double transgenic mice in which the loxP-flanked target gene will be deleted in the tissue where Cre is expressed.

In this example, Cre is expressed in the liver tissue so that deletion of the loxP-flanked gene occurs only in the liver of the double transgenic.

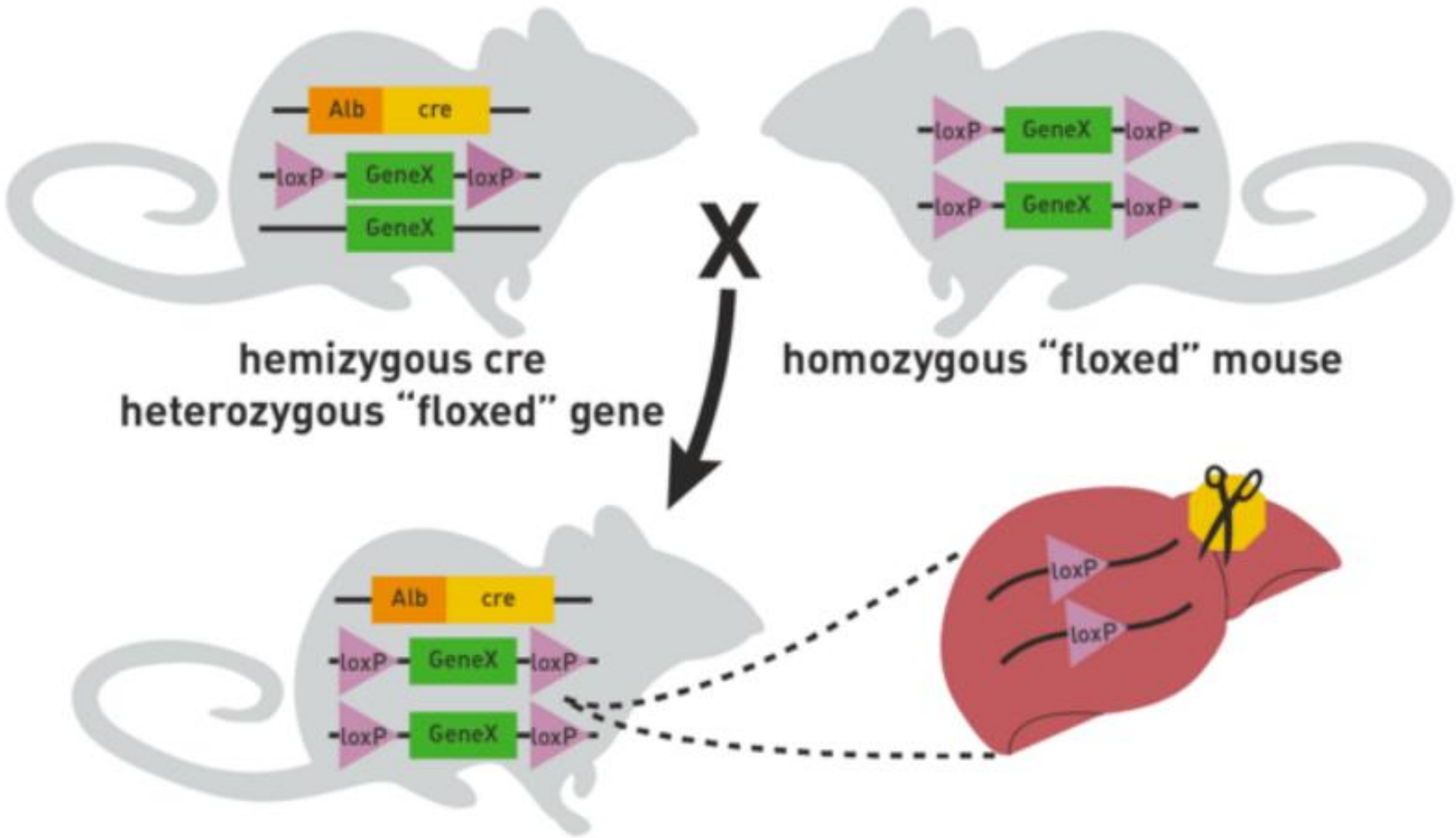
Other tissues and organs still express the loxP flanked gene.

Cre-lox *Tissue-Specific* Knockout



Cre-lox mouse:
heterozygous for GeneX conditional knockout after 1 generation

Cre-lox Tissue-Specific Knockout, cont.

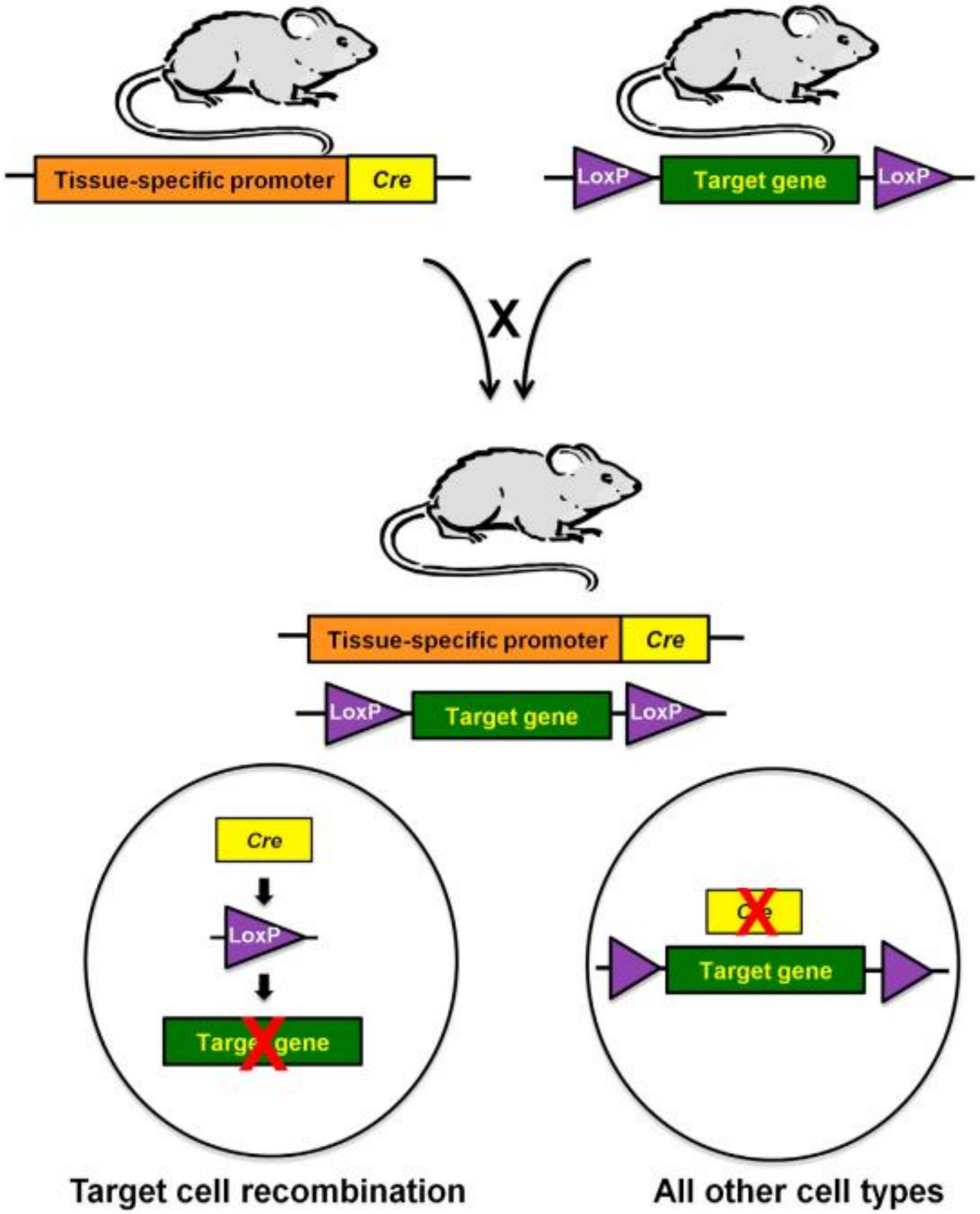


To generate mice that are heterozygous for a *loxP*-flanked allele and hemizygous/heterozygous for the *cre* transgene, mate a homozygous *loxP*-flanked mouse of interest to a cre transgenic mouse strain.

Approximately 50% of the offspring will be heterozygous for the *loxP* allele and hemizygous/heterozygous for the *cre* transgene.

Mate these mice back to the homozygous *loxP*-flanked mice.

Approximately 25% of the progeny from this mating will be homozygous for the *loxP*-flanked allele and hemizygous/heterozygous for the *cre* transgene. These will be your experimental mice.


Tissue-specific promoter
Cre
LoxP
Target gene
LoxP
X
Tissue-specific promoter
Cre
LoxP
Target gene
LoxP
Cre
LoxP
Target gene
Target gene
Target cell recombination
All other cell types

Somatic Cell Nuclear Transfer

Somatic Cell Nuclear Transfer (SCNT) is a powerful cloning technique that demonstrates how a fully differentiated cell can be reprogrammed to generate an entire organism.

SCNT is based on the principle that **all somatic cells retain a complete genome**, even after differentiation. What changes during differentiation is not DNA sequence, but **gene expression patterns** (epigenetic regulation).

The cytoplasm of an oocyte contains **reprogramming factors** capable of resetting a somatic nucleus to a **totipotent/embryonic state**.

Methodology

1. Donor Cell Preparation

- Somatic cells (e.g., fibroblasts) are isolated.
- Often synchronized in the **G0/G1 phase** to improve reprogramming efficiency

2. Oocyte Collection and Enucleation

- Mature oocytes are harvested.
- The nucleus (containing maternal DNA) is removed using micromanipulation.

3. Nuclear Transfer

Two approaches:

- **Direct nuclear injection**
- **Cell fusion (electrofusion)** – most common

The somatic nucleus is introduced into the enucleated oocyte.

4. Oocyte Activation

Artificial activation mimics fertilization.

Methods:

    Electrical pulses
    Chemical agents (e.g., calcium ionophores)

This triggers: Cell cycle resumption, Embryonic gene activation

5. Nuclear Reprogramming (Key Step)

The somatic nucleus undergoes extensive reprogramming:

- **DNA demethylation**
- **Histone modification changes**
- Resetting of genomic imprinting (often incomplete)

This converts:

Differentiated nucleus → Totipotent embryonic nucleus

6. Embryo Development

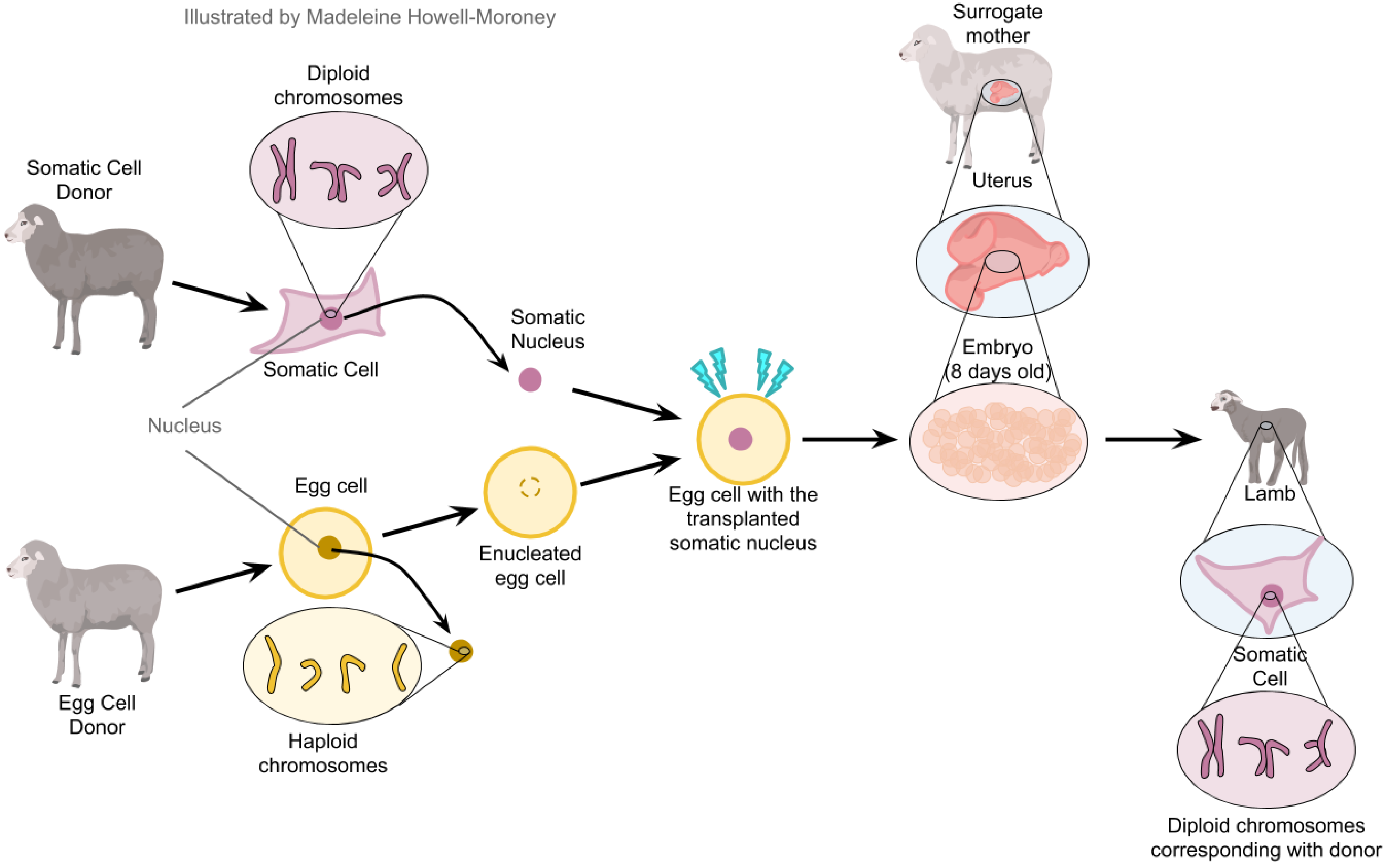
- Cleavage divisions occur → morula → blastocyst
- Blastocyst contains:
  - Inner Cell Mass (ICM)
  - Trophoblast

7. Implantation

- The embryo is transferred into a surrogate mother
- Development proceeds to term (if successful)

Somatic Cell Nuclear Transfer (SCNT)

Illustrated by Madeleine Howell-Moroney



Advantages

- Potential to be used to reproduce extinct species
- Retains genetic code of the donor nucleus
- Common procedures and method

Disadvantages:

- Often results in abnormal phenotypes in clones due to incomplete reprogramming
- Inefficient process with majority of studies reporting between 0.5 - 5% develop correctly

Landmark Example of SCNT

•Dolly the Sheep

- Created in 1996 by Ian Wilmut and team
- Proved that **adult somatic nuclei can be reprogrammed**

In 1996, cloning was revolutionized when **Ian Wilmut** and his colleagues at the Roslin Institute in Edinburgh, Scotland, successfully cloned a sheep named **Dolly**. Dolly was the first cloned mammal.

Wilmut and his colleagues transplanted a nucleus from a mammary gland cell of a Finn Dorsett sheep into the enucleated egg of a Scottish blackface ewe.

The nucleus-egg combination was stimulated with **electricity** to fuse the two and to stimulate cell division. The new cell divided and was placed in the uterus of a blackface ewe to develop. Dolly was born months later.

Dolly was shown to be genetically identical to the Finn Dorsett mammary cells and not to the blackface ewe, which clearly demonstrated that she was a successful clone (it took **276 attempts** before the experiment was successful). Dolly has since grown and reproduced several offspring of her own through normal sexual means. Therefore, Dolly is a viable, healthy clone.

Since Dolly, several university laboratories and companies have used various modifications of the nuclear transfer technique to produce cloned mammals, including cows, pigs, monkeys, mice and Noah.

Dolly

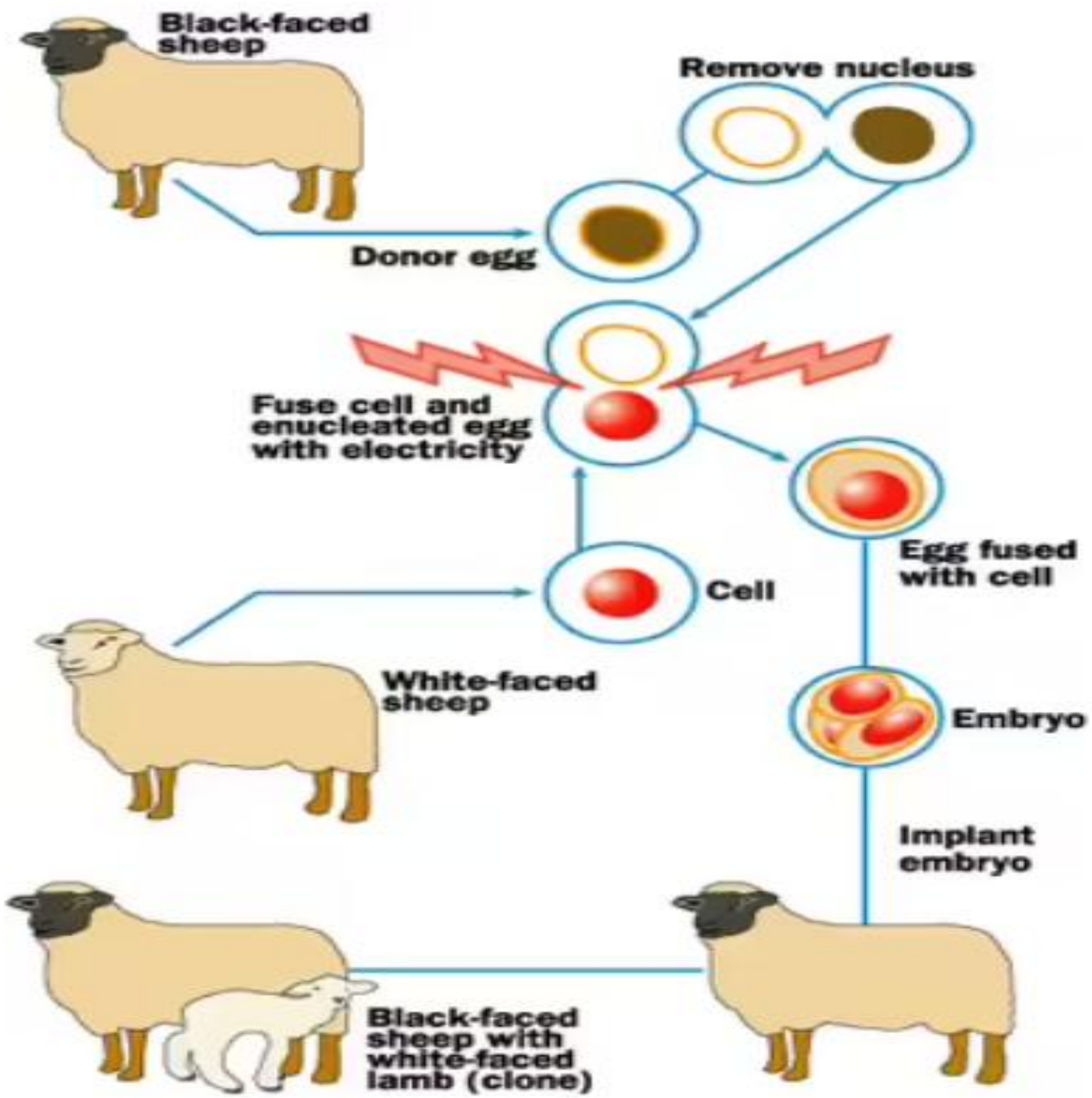


Diagram of the nuclear transfer procedure that produced the first cloned mammals

Sperm-Mediated Gene Transfer

Sperm-Mediated Gene Transfer (SMGT) is a technique for creating transgenic animals by using sperm as vectors to deliver exogenous DNA into oocytes during fertilization.

It involves incubating sperm with DNA, sometimes aided by techniques like electroporation or liposomes, followed by artificial insemination or IVF.

It is known for being simple and cost-effective, though often having lower efficiency.

Key Aspects of the SMGT Method

•**Mechanism:** Sperm cells naturally bind to exogenous DNA and transfer it to the oocyte during fertilization.

•**Enhancement Techniques:** To improve efficiency (which can be low), several approaches are used to aid DNA uptake by sperm, including:

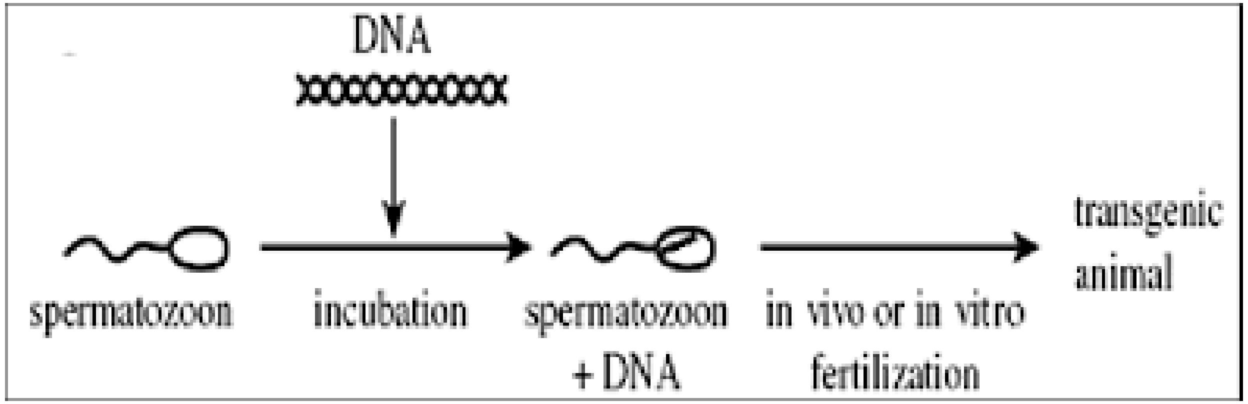
- **Electroporation:** Applying an electric field to make the membrane permeable.
- **Liposome-based methods:** Using liposomes to deliver DNA.

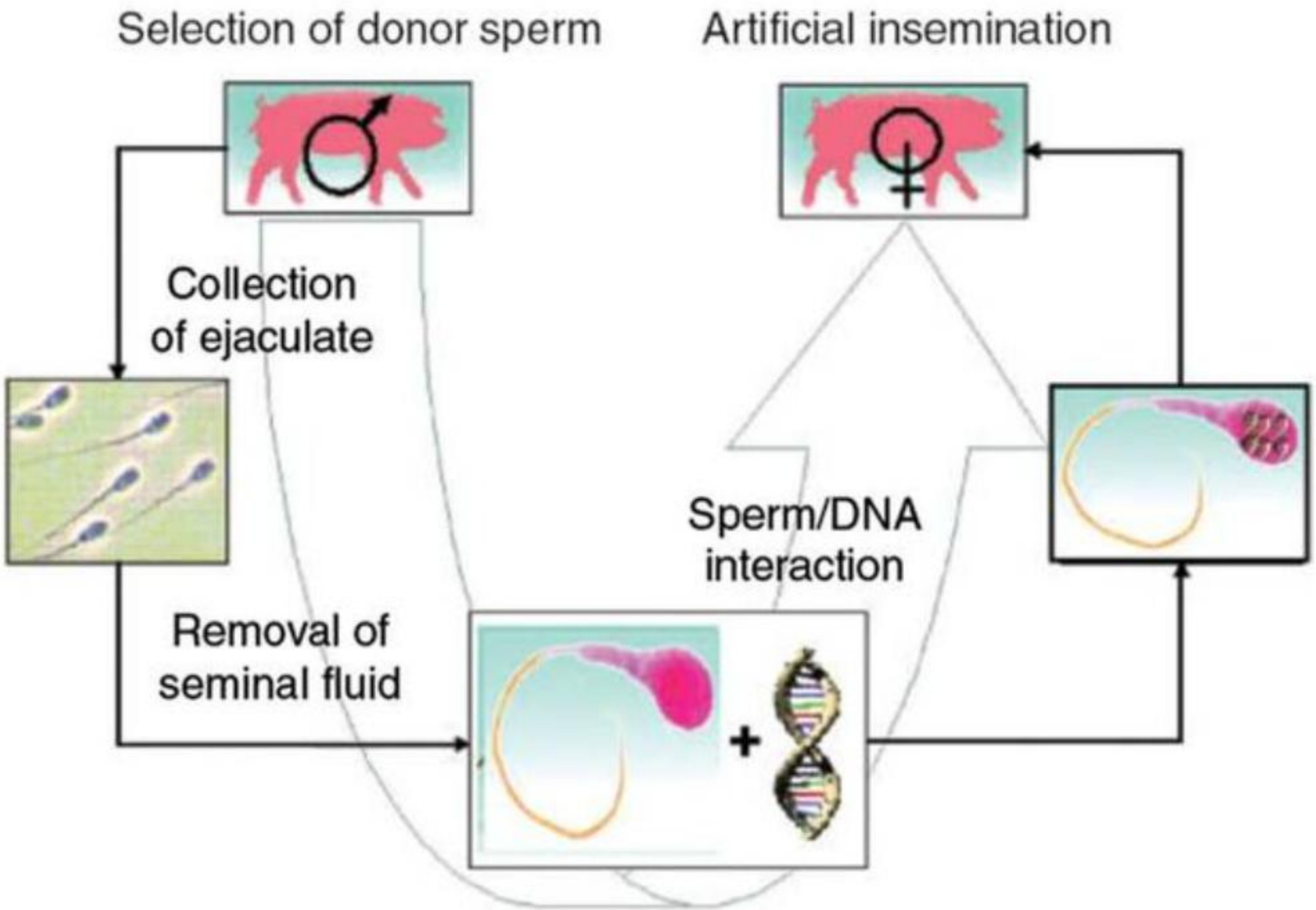
Procedure

1.DNA Incubation: Purified transgene DNA is incubated with sperm cells.

2.Fertilization: The DNA-bound sperm fertilizes the oocyte, typically via In Vitro Fertilization (IVF) or Intracytoplasmic Sperm Injection (ICSI).

3.Embryo Transfer: The resulting transgenic embryo is transferred to a recipient female.





Selection of donor sperm
Artificial insemination
Collection
of ejaculate
Sperm/DNA
interaction
Removal of
seminal fluid
+

Advantages:

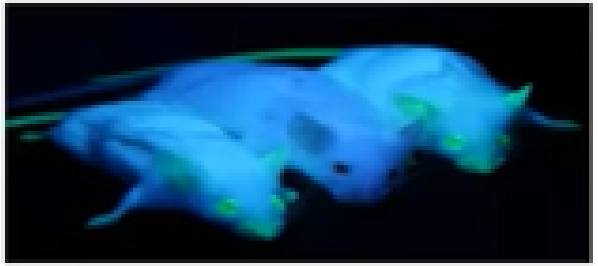
- **Low Cost & Simple:** It is simpler and often less expensive than traditional methods like embryo microinjection.
- **No Complex Equipment:** Generally requires less specialized equipment.
- **Versatility:** Successful in various species, including mice, pigs, goats, and cattle

Limitations:

- **Low Efficiency:** Transgene integration efficiency is often low and variable.
- **Variable Binding:** DNA binding capacity to sperm varies significantly among individuals and species (ranging from 0.3% to 78%).
- **DNA Degradation:** Spermatozoa have mechanisms (like nucleases) that can degrade the foreign DNA.

Transgenic Mice

- The first transgenic animals were mice created by Rudolf Jaenisch in 1974. He inserted foreign DNA into the early-stage mouse embryos resulting in mice carrying modified gene in all their tissues.



- **KNOCKOUT MICE-** widely used in medical research to investigate gene function, as they have their selected genes inactivated.
- They carry knock-out gene (non functional gene) at a place of interest. These are used for genetic analysis of inherited diseases and cancer.
- The transgenic mice were generated based on inheritance patterns of human disease. Therefore -
    - Introduction of collagen gene in mice genome produces knockout mice suffering Osteogenesis. It is, then, used to treat human diseases Osteogenesis.

Transgenic Mice

- **SUPERMOUSE** - was developed by Ralph Brinster (1982). It had the ability of superior learning and memory. It was used for meat production.
- By introducing Rat growth hormone gene in mouse, it resulted in:
  - high level of growth hormone.
  - Attains approximately double the weight of normal offspring mouse.

Oncomouse -

- 1st patented transgenic animal.
- It was developed to study Cancer and to screen anti – tumor drugs.
- It was made by inserting activated oncogenes.
- 13 different strains were engineered to contain human oncogene causing tumor formation.

Smartmouse -

- It was engineered to overexpress NR@B receptor in the synaptic pathway.
  - It increases the efficiency of mice to learn faster like juveniles throughout their life.

Transgenic livestock

- Microinjection is the most common gene transfer method for transgenic livestock. It has low efficiency as compared to transgenic mice.

Transgenic cow –

- ROSIE was the first transgenic cow, born in 1997
- Transgenic cows containing two types of casein gene produces 13% more milk.
- Transgenic cows with K-Casein Gene insert produces milk rich in K-Casein, which is further used for cheese making.
- Also, cattles with Lactose gene insert produces milk which is poor in lactose, that is useful for lactose intolerant people.
- Prion free transgenic cows are resistant to mad cow disease.

Transgenic livestock

Transgenic sheep –

- Tracy is the first transgenic animal to produce recombinant protein in its milk.
- It is used for good quality of wool production.
- It is used as a model for studying immunology, human blood clotting factor IX, transplantation, haematology, biological product manufacturing, recombinant DNA, drug production in milk.

Transgenic livestock

Transgenic goats –

- It is mainly developed for animal farming.
- Transgenic goats could express tissue plasminogen activator, anti thrombin III, spider silk etc in milk.
- tPA gene of plasminogen tissue also helps in dissolving blood clots.
- Silk gene are transferred from spiders, in order, to produce several grams of silk protein in her milk.
- Monoclonal Antibodies (MAbs) , Ig fusion proteins, tPA(tissue Plasminogen Activator), ATryn (recombinant human antithrombin III) is the first transgenic recombinant protein from transgenic goat approved by the United States Food and Drug Administration (USFDA) in January 2009.

Transgenic livestock

Transgenic goats –

- Disadvantages of transgenic livestock

Expensive.

Difficult procedure.

Transgenic Pigs

- Pigs are the only animal whose physiology matches to that of humans. Most common example of transgenic Pig is Enviropig.

- Normal pigs face trouble in fully digesting a compund, phytase, found in many cereal grains of their diet.While, transgenic pigs are created by introduction of phytase gene obtained from E.coli.
    - Phytase enzyme in salivary gland of transgenic pig aids in degradation of indigestible phytase by releasing phosphate. Pigs easily digest phosphates.
    - A genetically engineered pig approved for limited production which produces 65% less phosphorous in animal waste.
- Transgenic pigs are also used for Organ transplant harvesting.
- Transgenic pigs inserted with Human ⊠- globulin gene, produces human hemoglobin in blood. It is used to treat Haemophilia.

Transgenic Fish

- Transgenic fish are produced by the artificial transfer of genes into fertilized eggs.. Microinjection is the desired method, though the integration rates of transgenes are generally low.
- SUPERFISH - Salmon/trout, Tilapia, Catfish
  - Growth hormone (GH) gene are inserted into fertilized egg by microinjection method.
  - Therefore, have extra copies of GH.
  - Leads to increased growth and size, upto 6 times faster than control fish.

- GLOW FISH – Zebra fish (Danio carpio)
  - Ornamental GloFish are produced by integrating a fluorescent protein gene from jelly fish into embryo of a fish.
  - It is the first genetically modified animal to become publicly available as a pet (contain red, green, yellow and orange fluorescent color).

Applications of Transgenic Animals

Medical applications –

Disease model for humans for studying.

- Genetic basis of human and animal disease such as Cancer, Cystic fibrosis, Rheumatoid arthritis, Alzheimer's etc.
- Disease resistance in humans and animals.
- Drug and product testing and/or screening.

Xenotra production of pharmaceutical in transgenic animals

- Transgenic cows, sheep and goats are able to provide nutritional supplements such as insulin, growth hormone and blood clotting factors.

Transplantation

- Patients die every year due to lack of organs for replacement (heart, liver, or kidney).

Applications of Transgenic Animals

Agricultural applications –

Breeding -.

- Transgenesis had made it possible to develop desired traits such as increased milk production, high growth rate etc by selective breeding in animals in a shorter time with more precision. It also increases the yield production.

Quality -

- HERMAN, a transgenic bull, was developed that carries a human gene for lactoferrin (increases iron content).
- Transgenic pigs and cattle weigh approximately double than normal animals and are used for meat production.

Applications of Transgenic Animals

Industrial applications -

The extraction of polymer strands from the milk helped the scientists to create light, tough, flexible material that are used further for weaving military uniforms, medical micro sutures, tennis racket strings etc.

To produce wide variety of enzymes, micro-organisms are engineered through transgenesis, which are also capable of producing enzymes that can speed up industrial chemical reactions.

Ethical Concerns

- Genetically modified animals have been intentionally designed to suit human purposes using modern technology, but use of animals in science had always been controversial issue.
- Some of the progress are considered morally unacceptable e.g. Cloning.
- There are some moral or ethical factors concerned to genetic manipulation.

Principle of 3 R's –

- Reduction of animal numbers.
- Refinement of practices and husbandry to minimize pain and distress.
- Replacement of animals with non – animal alternatives wherever possible.

Ethical Concerns

Animal welfare concerns –

- Use of animals in research causes great suffering to the animals.
- Each biological species has a right to exist as a separate identifiable entity.
- Animals should have the same basic rights as human beings.
- Due to experimentation, it had diminished the sense of identity and individuality.

- Xenotransplantation Issues –transplantation or organs from transgenic pigs to humans can spread Zoonotic disease such as Mad Cow disease, that could have devastating consequences.

Production of vaccines in animal cells

Production of vaccines in animal cells refers to using cultured mammalian cells to grow viruses or express antigens for vaccine manufacture. This approach is widely used for viral vaccines and recombinant protein vaccines.

General Process:

1. Selection of cell line

Commonly used cell lines:

- **Vero cells** (monkey kidney cells)
- **CHO cells (Chinese Hamster Ovary)**
- HEK293 cells

2. Antigen/virus production

Two approaches:

- **Live virus cultivation** → cells infected with virus (e.g., polio, rabies)
- **Recombinant protein expression** → cells engineered to produce antigen proteins

4. Harvesting

- Virus or antigen is collected from:
    - Cell lysate or
    - Culture medium

5. Purification

- Removal of impurities using:
    - Filtration
    - Chromatography

6. Inactivation/attenuation (if required)

- For inactivated vaccines:
    - Chemicals like formaldehyde used

7. Formulation

- Mixed with:
    - Adjuvants
    - Stabilizers
- Final vaccine is prepared

Examples of vaccines produced in animal cells

•**Polio vaccine (IPV)** – Vero cells

•**Rabies vaccine** – Vero or human diploid cells

•**Influenza vaccines** – MDCK cells

•**Hepatitis B vaccine** (recombinant, though often yeast-based also)